

12.- Sample Inspection IBDs

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1 Introduction:

IBDs need to be set up in MADRAS at sample level, prior to start the Sample Change Control Program (first step of Sample Inspection)

IBD's are basically MBD's set up for purposes of sample inspection. IBD definition needs to be decided by the Local Stat Ops Team and Local DSci Market Leaders (Client Engagement)

2 Process to be Followed:

2.1 General recommendation for IBD:

1. Every store in the sample needs to belong to one and only one IBD per period. This means we cannot have IBDs overlapped.
2. IBD's are cell-based MBD's.
3. IBD's should have a minimum number of stores to ensure the robustness of the computations (standard errors, interval of confidence...). The minimum number recommended is 30 stores, although we can permit less stores in some special situations where to group the stores with others in another IBD has no so much sense (for instance the extreme case of mixing Hypermarkets with Traditional stores in some specific region). The max recommended number of stores to be handled should be around 100. (Due limitation of current Sample Inspection tool in Madras, to avoid very long executions, and not to jeopardize production cycle)
4. They must be (IBDs) market segments linked to a unique sample.
5. IBDs should be homogeneous based on basket sales.
6. There are countries that have in the same sample, some cells that are expanding with X_factor, and other cells that are expanding with Z_factor. In this case, these cells can't be in the same IBD. All the stores belonging to the same IBD must expand with the same type of exp factor (X or Z).

2.2 Homogeneity in Sales:

One of the recommendations of IBD setup is to have homogenized basket sales. In order to evaluate it, we could use ANOVA test. Analysis of variance (ANOVA) test is used to analyze the differences among group means in a sample. DS Design Center to evaluate what tool or analysis, ANOVA is just a recommendation

Ideal V value ~ 50%

For country with small sample or having less IBD group, may consider using Coefficient of Variation (CV) instead.

$CV = \sqrt{\text{Var Sales}} / \text{Avg Sales}$

In case the CV of an IBD is extremely high, it's important to review if there is a need to combine or split the IBD grouping, as high CV indicate that the variation of sales within the IBD is huge.

2.3 Special recommendation:

Based in experience, we could recommend as initial point to start IBD's by Shoptype/NIQ Region when sufficient sample, as sample design must considered the sales homogeneity and should also be cell based.

3 Special consideration to accelerate sample inclusion

3.1 When country is in a sample enlargement process:

We know that NIQ is aiming to enlarge current audit samples (traditional shootypes/markets) with as many as possible stores coming from POS (scanning system) either Audit to Scan, TTR, ...

Here the main consideration will be to have as more IBD's possible, to allow more stores to be incorporated each time, still under the criteria to minimize the impact at IBD level. Again, we should have IBDs with more than 30 stores & less than 100

In those cases:

1. IBD should not have data from a mixed time frame - IBD should not contain data for example monthly collected shops and weekly POS shops. In the same way they should not be under same Cell and Sample (Sample only can have stores from one periodicity)
2. IBD should not have mixed data from Scan and Audit. Even if same periodicity (this is not a must, but strongly recommended)
3. In case that with the enlargement of sample we will pass the 70 stores we should look if there is a good option a possibility committing all previous rules for splitting those IBD's to speed the sample inclusion process.

4 Queries to check IBDs

To review IBDs you could follow next document that also include some queries to support us

[Queries_to_check_IBDs.docx](#)¹

1. https://adlm.nielseniq.com/confluence/download/attachments/534454784/Queries_to_check_IBDs.docx?api=v2&modificationDate=1689837153572&version=1